

Dylan Dsouza

San Diego, California

✉ dydsouza@ucsd.edu

🌐 dylansouza.com

🌐 [linkedin.com/in/dsouza-dylan](https://www.linkedin.com/in/dsouza-dylan)

🐙 github.com/dsouza-dylan

EDUCATION

University of California San Diego (Provost Honors)

Bachelor of Science in Data Science (GPA: 3.95)

San Diego, CA

Expected Mar 2027

TECHNICAL SKILLS

Languages: Python, R, SQL (PostgreSQL), Java, HTML, CSS, JavaScript

Libraries/Frameworks: PyTorch, TensorFlow, pandas, scikit-learn, statsmodels, OpenCV, NLTK, LangChain, Neo4j, Jupyter, Git

Core Competencies: Data Cleaning/Visualization, EDA, ETL Pipelines, A/B Testing, Financial Modeling, Time Series, NLP

EXPERIENCE

Machine Learning Researcher

Oct 2025 – Present

Engineers for Exploration

San Diego, CA

- Collaborated on a DeepLab-powered Model-Assisted Labeling pipeline that processes 21,000+ GeoTIFF image-mask pairs to automate mangrove annotation and segmentation, with outputs deployable in ArcGIS for use by field researchers.
- Integrated 7 geospatial datasets (SpaceNet, xView, LandCoverAI) to train ensemble segmentation models for pixel-level mangrove mapping, developing gated inference architecture for land-cover classification across 4 environmental classes.

AI Research & Development Intern

Oct 2025 – Present

Computer Science & Engineering Society

San Diego, CA

- Engineered a 3-stage hybrid RAG retrieval pipeline using LangChain and Neo4j, combining vector search, document-aware MMR, and cross-encoder reranking with graph-based context expansion to improve retrieval for an agentic tutoring platform.
- Built a knowledge graph pipeline that extracts LLM-generated concepts from research documents, mapping relationships between them using contextual proximity scoring and Girvan-Newman community detection across 8+ entity categories.

Research Data Analyst

May 2025 – Present

Scripps Institution of Oceanography

San Diego, CA

- Automated 11 years of coral reef data processing (60% cut in manual processing time), building a Python ETL pipeline that standardized 40,000+ annotations into clean datasets for downstream analysis across 15 reef sites.
- Tracked coral ecosystem response to a 2023 mass bleaching event across 5 treatment groups, applying ANOVA, Tukey HSD, and LOWESS-smoothed time series analysis to reveal statistically significant recovery differences across all functional groups.

Quantitative Analyst

Oct 2025 – Dec 2025

Sustainable Investment Group

San Diego, CA

- Built an automated Python pipeline using FinBERT and Yahoo Finance API to quantify lagged correlations between 25 weeks of news sentiment and 5+ market KPIs, generating actionable trading signals from real-time financial data.
- Co-led quantitative investment thesis for Under Armour integrating ARIMA forecasting, FinBERT sentiment analysis, and DCF modeling to deliver a sell recommendation across a 5-team competitive stock pitch.

Market Research Intern

Jul 2023 – Aug 2023

Bennett Coleman & Co. Ltd. (The Times of India)

Mumbai, India

- Built a multiclass NLP classifier achieving 85% accuracy across 5 market sectors, processing 56,000+ newspaper headlines using TF-IDF and BERT embeddings to automate market sector categorization at scale.
- Researched enterprise data infrastructure modernization strategies, including data lakes, CDPs, and CRM optimization, as part of a 5-person team to support internal evaluation of workflow improvements across editorial and business operations.

PROJECTS

NextMove: Predicting Next Workout Activity

Nov 2025 – Dec 2025

Course Project — Recommender Systems

[View Project](#)

- Engineered temporal, behavioral, and physiological features from 20,000+ sequential workout records, designing transition-based training instances to predict next workout activity without look-ahead bias.
- Benchmarked 3 heuristic baselines (frequency, recency, weighted) against supervised ML models (Logistic Regression, Naïve Bayes, SVM, Random Forest), achieving 87% accuracy with Random Forest using chronological splits and balanced classes.

ENSOCast: Interactive Climate Analytics Dashboard

May 2025 – Oct 2025

Personal Project and Conference Presentation

[View Project](#)

- Presented at the 2025 San Diego Undergraduate Tech Conference; built an interactive Streamlit dashboard exploring 40+ years of ocean temperature data for El Niño-Southern Oscillation (ENSO) phase analysis.
- Developed and benchmarked 5 ML models (Random Forest, XGBoost, 1D CNN, LSTM, Ensemble) using temporal and seasonal features, achieving 82% accuracy in ENSO phase prediction.

[Portfolio](#)

[Additional Projects](#)

[GitHub](#)